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Undergraduate students' learning outcomes with ChatGPT: A meta-analytic study

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ABSTRACT

ChatGPT has gained substantial attention in the field of higher education, particularly for its potential to enhance undergraduate students' learning outcomes. To better understand ChatGPT's impact, we conducted a meta-analysis evaluating the effects of ChatGPT applications on undergraduate students' learning outcomes, with data collected from studies published between January 1st, 2023, and May 31st, 2025. Our search across nine academic databases identified 5555 potential studies, of which 66 met the pre-defined inclusion criteria and were selected for meta-analysis. The meta-analysis incorporated 129 effect sizes, allowing us to estimate the overall impact of ChatGPT on undergraduate students' learning across a variety of academic disciplines. The results suggested that ChatGPT applications had a large positive effect (Hedges' $g = 1.14$, $SE = 0.185$) on undergraduate students' learning outcomes. The results of this study highlight undergraduate students' overall positive experiences with ChatGPT. These findings contribute to the growing body of literature on the role of artificial intelligence (AI) in higher education, offering critical insights for educators, administrators, and policymakers seeking to enhance undergraduate students' learning outcomes by integrating AI technologies like ChatGPT into academic curricula.

1. Introduction

Since its emergence, ChatGPT has attracted significant attention for its potential to revolutionize various aspects of education and learning (Baidoo-Anu & Ansah, 2023). However, the rapid integration of ChatGPT into educational environments has also sparked a heated debate within the academic community (Coeckelbergh & Gunkel, 2024; Jiang et al., 2024). The core of this debate lies in the mixed findings reported in the literature. While some studies highlight the positive impact of ChatGPT on student engagement, comprehension, and overall learning outcomes (Songsiengchai et al., 2023; Yıldız, 2023), others raise concerns about its effectiveness, questioning whether the benefits are consistent across different educational contexts or if they come with unintended drawbacks (Uzun, 2023; Wu et al., 2024). This debate is fueled by the variability in study designs, sample sizes, and methodologies across existing research, leading to a lack of consensus on the true impact of ChatGPT in educational settings. As a result, educators, administrators, and policymakers are left with conflicting information, making it challenging to make informed decisions about the integration of AI technologies like ChatGPT into curricula.

Given this context, meta-analysis emerges as a particularly valuable approach for addressing these uncertainties. Unlike individual studies, which might focus on specific contexts or outcomes, a meta-analysis systematically synthesizes empirical quantitative findings from multiple studies, allowing for a more comprehensive and statistically robust assessment (Borenstein et al., 2021; Shuster, 2011) of ChatGPT's impact on undergraduate students' learning experiences. By pooling data from various sources, meta-analysis increases the overall sample size and statistical power, providing more precise estimates of the effect size (Cooper et al., 2019). This approach also allows for the exploration of heterogeneity, helping to identify factors that might explain why the impact of ChatGPT varies across different studies (Borenstein et al., 2021). Moreover, meta-analysis can mitigate biases present in individual studies, such as publication bias, where only studies with significant

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results are published, while those with null or negative results remain unpublished (Rosenthal, 1979). By incorporating all available evidence, meta-analysis offers an unbiased view, making it a powerful tool for evidence-based decision-making.

This study seeks to provide empirical evidence by synthesizing the growing body of studies on the impact of ChatGPT on undergraduate students' learning outcomes. Through meta-analysis, the study aims to facilitate a more informed and evidence-based discussion on the effective integration of Artificial Intelligence (AI) technologies in higher education, addressing the ongoing debate with a comprehensive and rigorous evaluation of the available data.

2. Literature review

2.1. Brief history of AI in education

AI applications in education first emerged with the rise of computer-assisted instruction in the 1960s and have since made considerable progress with the development of advanced technologies (Woolf, 2010). The widespread use of internet-based learning and data-driven educational tools has set the stage for AI to play a more integral role in various educational settings (Zawacki-Richter et al., 2019). Adaptive learning platforms have transformed education by personalizing student experience, employing big data to optimize learning paths based on real-time feedback and performance analytics (Roll & Wylie, 2016). These tools personalize the learning experience, adjusting to individual student's needs, thereby enhancing learning outcomes (Kulik & Fletcher, 2016). Later, the introduction of large language models (LLMs) like ChatGPT has significantly expanded AI's potential in education (Jiang et al., 2024), unlocking new possibilities for personalized learning with tailored feedback, flexible learning pace, and access to a wide range of information (Firat, 2023; Haleem et al., 2022). This made it also possible to address a wide spectrum of educational needs and learning styles with unprecedented flexibility (Limo et al., 2023).

2.2. Applications of ChatGPT among undergraduate students

Since its initial launch in November 2022, ChatGPT has rapidly gained prominence in multiple sectors, including higher education (Dempere et al., 2023; Michel-Villarreal et al., 2023). Its adoption in educational settings is driven by its potential for personalized learning (Limo et al., 2023), and its role in enhancing student engagement (El-Seoud et al., 2023). ChatGPT's human-like text generation capabilities have made it increasingly popular among undergraduate students for various learning purposes (Castillo et al., 2023; Lo, 2023). ChatGPT has been applied in diverse disciplines in higher education, including business (George & George, 2023), computer science (Bordt & von Luxburg, 2023), engineering (Zhang et al., 2023), gastroenterology (Klang et al., 2023), healthcare (Muftić et al., 2023), languages (Song & Song, 2023). The wider adoption of AI technologies in education, exemplified by ChatGPT, is also promoting the shift from traditional student interactions with learning materials and course content to innovative, technology-mediated approaches (Grassini, 2023).

2.3. Current applications of ChatGPT in higher education

The integration of artificial intelligence (AI) in higher education has transformed teaching and learning methodologies, with ChatGPT at the forefront due to its advanced language processing capabilities. A range of studies, including those by Zhu et al. (2023), demonstrate how ChatGPT enhances student engagement and facilitates collaborative learning, thereby improving learning outcomes across disciplines. In language education, Muniandy & Selvanathan (2025) and Baskara (2023) have shown that ChatGPT significantly aids English as a Second Language (ESL) and English as a Foreign Language (EFL) learners by providing interactive language practice, which boosts fluency and engagement. Moreover, ChatGPT's application in academic writing and feedback, highlighted by Escalante et al. (2023), offers students instant, actionable insights that enhance their writing skills. Additionally, Pardos and Bhandari (2024) explored ChatGPT's effectiveness in tutoring and problem-solving within complex subjects like mathematics, comparing favorably to traditional human tutoring. Finally, Wang and Feng (2023) and Yilmaz and Yilmaz (2023) illustrate how ChatGPT can be integrated into curricula to develop educational content and engage students more deeply with course materials. These studies collectively underscore ChatGPT's versatile role in revolutionizing educational experiences, suggesting a promising future for AI in education.

2.4. Controversial sentiment towards ChatGPT and AI applications in education

While AI technologies such as ChatGPT offer great benefits in educational settings, there is still a degree of controversy and mixed sentiment regarding the use of ChatGPT and other AI technologies in education (Rejeb et al., 2024). First, ChatGPT may provide inaccurate information, which could negatively affect students' learning (Johnson et al., 2023). Secondly, there are risks associated with exposing personal information to unauthorized parties, raising privacy and security issues (Wu et al., 2024). Thirdly, not all students have equal access to AI technologies, which could widen the digital divide and exacerbate existing educational inequalities (Khan & Paliwal, 2023). Fourthly, dependence on AI may hinder the development of critical thinking and problem-solving skills (Al-Zahrani, 2024). Additionally, over-reliance on ChatGPT can make students less inclined to seek information independently and engage deeply with the material (Kosmyna et al., 2025). Furthermore, AI tools may lead to academic dishonesty, such as plagiarism or cheating (Huallpa, 2023). Lastly, ChatGPT may reinforce existing biases, perpetuating stereotypes and misinformation (Farhi et al., 2023). Without proper context or critical thinking skills, students might be influenced by biased or incorrect information, which could negatively impact their understanding and perpetuate a harmful stereotype in educational settings (Busker et al., 2023; Gross, 2023).

The ongoing debate underscores the need for a balanced approach that maximizes AI's educational potential while mitigating associated risks (Zawacki-Richter et al., 2019). This meta-analysis could contribute to a better understanding of these concerns by providing empirical evidence on the functions and effects of ChatGPT. It could also offer a clearer view of the situation, helping to shape future developments and policies in the integration of AI in education.

2.5. Purpose of this study

The period following ChatGPT's introduction in November 2022 has been marked by an explosion of scholarly inquiry, resulting in vast and rapidly growing literature on its role and effects in education. This surge in research necessitates a comprehensive synthesis to understand the strengths and weaknesses of ChatGPT in these contexts. As highlighted by Baidoo-Anu and Ansah (2023), the accumulation of such a diverse array of studies calls for an organized effort to synthesize, analyze, and interpret these findings. The existing systematic reviews on ChatGPT in education have certain limitations, particularly in terms of the scope they encompass (Montenegro-Rueda et al., 2023; İpek et al., 2023). They have focused on the initial months following ChatGPT's launch, providing a limited perspective on its long-term implications in educational settings. This meta-analysis aims to fill this gap by delving into studies published between January 1st, 2023, and May 31st, 2025, with a specific focus on the integration of ChatGPT in undergraduate students' learning experiences. By critically evaluating a broader range of recent studies, this study aims to provide a comprehensive understanding of ChatGPT's role and potential in enhancing undergraduate students' educational outcomes by illustrating a clearer picture of ChatGPT's effectiveness and areas for improvement in undergraduate students' learning. By analyzing these studies, we can provide critical insights for developing informed policies and strategies for the effective use of ChatGPT in educational contexts. This study is crucial for educators and policymakers to optimize the use of ChatGPT as a supportive educational tool, enhancing undergraduate students' learning and addressing potential challenges. It also contributes to the body of research by offering a better understanding of the current state of research regarding ChatGPT's effectiveness and provides valuable insights for future research on this topic.

The research questions designed to guide this study were.

RQ 1: What is the impact of ChatGPT applications on undergraduate students' learning outcomes?

1.1 What is the overall effect size, representing the effectiveness of ChatGPT application on undergraduate students' learning outcomes?

1.2 To what extent do effect sizes vary across studies?

RQ 2. To what extent do application contexts in which ChatGPT is applied moderate its impact across empirical studies?

2.6. Theoretical framework: Integrative AI in education (IAIE) framework

The Integrative AI in Education (IAIE) Framework, developed by synthesizing existing theories and models from literature, provides a comprehensive approach to understanding the impact of AI on education. Drawing from frameworks proposed by Guan et al. (2020), Hwang et al. (2020), Jantakun et al. (2021), and Langley (2019), and the Explainable AI in Education framework by Khosravi et al. (2022), this new model conceptualizes AI integration in education at three levels: learner-oriented, where AI tools support personalized learning experiences (Hwang et al., 2020; Khosravi et al., 2022); instructor-oriented, where AI aids educators in customizing teaching strategies and monitoring student progress (Hwang et al., 2020; Langley, 2019); and institutional system-oriented, where AI informs policy-making and optimizes educational environments (Guan et al., 2020; Jantakun et al., 2021; Ouyang & Jiao, 2021). AI's roles within this framework include acting as an intelligent tutor providing customized instruction, a learning partner assisting in complex cognitive tasks, and a policymaking advisor helping to shape educational strategies (Hwang et al., 2020). Moreover, the IAIE Framework emphasizes the importance of creating adaptive, transparent, and ethically sound learning environments. It ensures that AI tools are not only effective in personalizing learning but are also accessible, equitable, and used responsibly. Ultimately, the IAIE Framework offers a structured foundation for exploring AI's potential to transform education and enhance student learning outcomes, while also addressing key ethical and practical challenges.

3. Method

3.1. Study identification

The targeted studies for this meta-analysis are empirical quantitative studies that report the impact of ChatGPT use on undergraduate students' learning outcomes. We searched for such studies reported between January 1st, 2023, and May 31st, 2025, from nine electronic databases, including *APA PsycArticles, Educational Resources Information Center (ERIC), Education Full Text, Education Source, Google Scholar, ProQuest, PsycINFO, SCOPUS*, and *Web of Science*. We employed Boolean operators in a search strategy with keywords to optimize the search process. The search string, with some adjustments required for a particular database, we used was: ChatGPT AND (Intervention OR Empirical OR Experiment OR Learning OR Instruction) AND (College OR Higher education OR Undergraduate). We defined the inclusion/exclusion criteria to reflect the scope of the study (see Table 1), which resulted in identifying a total of 5554 potential studies for meta-analysis. After removing 2376 duplicates, 3178 articles remained for abstract screening. We then reviewed these abstracts and retained 607 articles for full text reviews by the first four authors, resulting in 540 articles being further excluded for various reasons. Ultimately, we had 65 articles for inclusion. After reviewing the references of these studies, we included one additional relevant study, bringing the total to 66 articles with 129 effect sizes from 5708 cases for synthesis. Fig. 1 illustrates the flow chart of the record selection process based on the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) 2020 guideline (Page et al., 2021). The screen process was managed with Rayyan (2016).

Table 1
Inclusion and exclusion criteria for article selection.

Criterion	Inclusion	Exclusion
Research Topic	Related to undergraduate students' learning experience with ChatGPT across different subject domains	Investigate topics other than undergraduate students' learning experience with ChatGPT
Study Design	Empirical studies	Other studies (reviews, meta-analysis, opinion articles, letters to editor, etc.)
Article Type	Primary studies in peer-reviewed journals, unpublished dissertations, conference papers	Systematic reviews, meta-analysis articles, books, magazine and newspaper articles, online sources (e.g. blogs)
Publication Period	From January 1st, 2023 to May 31st, 2025	Articles outside of this time frame
Population	2-year or 4-year full-time undergraduate students	K-12 students, part-time undergraduate students, graduate students, post-secondary students
Language	English	Written in languages other than English

3.2. Coding procedures

To develop the coding sheet for this meta-analysis, we followed a systematic and structured approach to ensure that all relevant data from the included studies could be accurately and consistently extracted. First, after determining the fundamental details of the study we extract, we identified the key variables to be investigated as potential moderators. To identify potential moderators, we selected variables that captured the diverse contexts and methodologies present in the primary studies. We resolved discrepancies through discussion over a series of meetings until we finalized the variable groupings.

Table 2 summarizes the coded key variables extracted from the 66 primary studies, which encompassed “Study Type”, “Majors”, “Previous Exposure to ChatGPT”, and “the Presence or Absence of Guidance in ChatGPT Application”. Students' prior experience with ChatGPT may influence how they interact with this tool. Ammari et al. (Ammari et al., 2025) 's study describes that students' accumulated experiences with the AI tool reflect their diverse usage patterns. By including previous experience as a moderator, the analysis can determine whether the effectiveness of ChatGPT on learning outcomes varies depending on students' familiarity with the technology.

We also considered different academic disciplines that may interact with AI tools like ChatGPT in varied ways (Wu & Zhang, 2023). For instance, students in the humanities might use ChatGPT for generating ideas or analyzing text, while STEM students might use it for coding help or solving technical problems. The relevance and utility of ChatGPT could differ across majors, potentially leading to varying effects on learning outcomes. By examining student major as a moderator, the analysis can explore whether the impact of ChatGPT differs by field of study. Finally, the way students are introduced to and guided in using ChatGPT can have a substantial impact on its effectiveness. Structured guidance, such as instructions on how to critically evaluate ChatGPT's

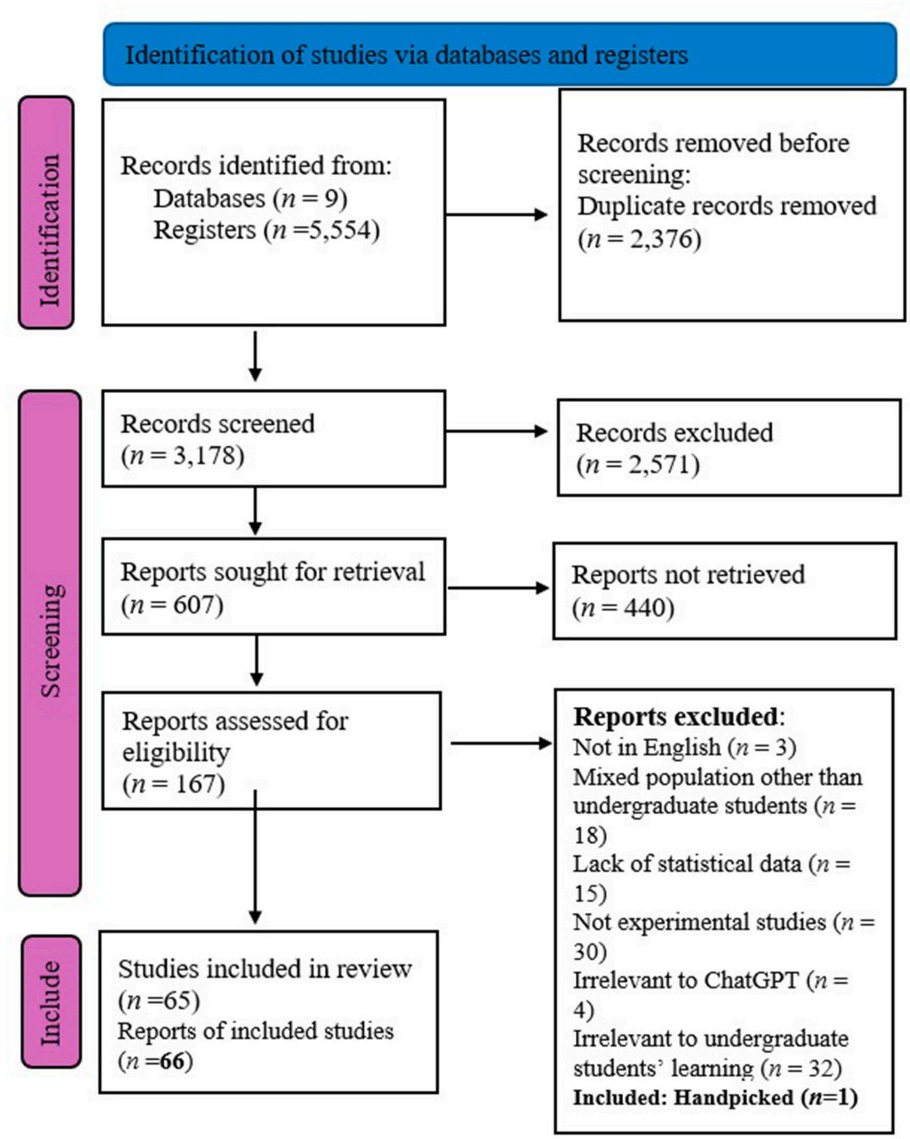

Fig. 1. Summary of study search procedures (adopted from the PRISMA Flowchart 2020).

outputs, could enhance learning outcomes by helping students use the tool more effectively and avoid potential pitfalls like relying on inaccurate information.

We established operational definitions for each variable to ensure consistency in how data were coded across different studies. For example, "Previous Exposure with ChatGPT" was defined in terms of the frequency of students' usage of ChatGPT prior to the study. Based on the defined variables, we developed specific coding categories. These categories included both general information about each study and more specific details related to the outcomes and moderators of interest. Some of the key coding categories included: Study Characteristics (e.g., authors, year of publication, country of study, study design, etc.), sample characteristics (e.g., sample size, major, etc.), intervention details (e.g., how ChatGPT was used, whether guidance was provided on its usage, etc.), and outcomes (e.g., specific learning outcomes, effect sizes, statistical significance, etc.).

We used MS Excel spreadsheet for coding to facilitate easy data entry and analysis. Before applying it to all included studies, we piloted the coding sheet on two studies. During this phase, the first three authors independently coded the same two studies and then compared their results, achieving an inter-rater reliability of above 90 %. Any inconsistencies that arose were resolved through discussion, which allowed the authors to refine the coding criteria and processes. Based on the feedback from the pilot, adjustments were made to the coding sheet to improve clarity and accuracy. After interrater reliability was established with the revised coding schemes, the authors divided the primary studies among them equally. In cases where coding difficulties arose, the authors cross-checked each other's work for accuracy. Regular research meetings were held to discuss and resolve any ambiguities or issues through consensus among the authors.

3.3. Effect size calculation

The selected articles varied significantly in their methodologies and outcomes with different metrics the primary studies employed. To address this challenge and ensure consistency, we decided to use the standardized mean difference with sample size adjustments, Hedges' *g* (Hedges, 1981), as a common metric of the effect size in our analysis. This choice was made because Cohen's *d* is upwardly biased in small samples, where Hedges' *g* applies a correction to reduce this small-sample bias (Borenstein, 2009). When effect sizes were not reported in the study, we calculated effect sizes from the available descriptive statistics and then converted them to Hedges' *g*.

Table 2
Table for the primary studies and variables employed to calculate effect sizes.

Study	Country	Study Type	Exposure	Majors	PAGCA	N	N_1	N_2	Learning Outcome	d	var_d	g	var_g
Abdulla et al. (2024)	Oman	2	4	2	1	26	13	13	Computer Programming in Class Exam1	1.623	0.205	1.648	0.198
		2	4	2	1	26	13	13	Computer Programming in Class Exam2	−0.108	0.154	−0.104	0.149
		2	4	2	1	26	13	13	Computer Programming in Lab Exam1	1.348	0.189	1.369	0.183
		2	4	2	1	26	13	13	Computer Programming in Lab Exam2	1.098	0.177	1.112	0.171
Ali et al. (2025)	Saudi Arabia	2	1	1	1	100	50	50	English Grammar Skills	11.78	0.734	11.69	0.728
Almineeai et al. (2025)	Saudi Arabia	2	4	1	1	18	18	18	Speaking Proficiency	0.282	0.113	0.275	0.111
Alshammari (2025)	Saudi Arabia	1	4	2	1	54	27	27	Learning Effectiveness	1.41	0.092	1.39	0.091
Alwasidi and Al-Khalifah (2025)	Saudi Arabia	1	3	2	2	52	52	52	Writing Proficiency - Details/ Examples	1.12	0.051	1.112	0.05
		1	3	2	2	52	52	52	Writing Proficiency - Mechanics	0.412	0.04	0.409	0.04
		1	3	2	2	52	52	52	Writing Proficiency - Organization	0.865	0.046	0.858	0.045
		1	3	2	2	52	52	52	Writing Proficiency - Top Sentence	1.153	0.051	1.144	0.051
		1	3	2	2	52	52	52	Writing Proficiency - Word Count	−0.394	0.04	−0.391	0.04
Ameen et al. (2024)	Iraq	1	4	3	1	146	46	48	Chemistry Achievement	1.283	0.048	1.273	0.048
		1	4	3	1	146	46	48	Chemistry Computational Thinking	0.855	0.045	0.848	0.045
		1	4	3	1	146	25	27	Computer Science Achievement	2.339	0.096	2.303	0.094
		1	4	3	1	146	25	27	Computer Science Computational thinking	2.289	0.095	2.254	0.094
Arkan et al. (2025)	Türkiye	1	3	2	2	96	48	48	Nursing Process Competency	1.24	0.05	1.22	0.049
		1	3	2	2	96	48	48	Problem-solving Skills	0.6	0.044	0.59	0.043
Barreto and Abarca (2025)	Peru	1	1	2	1	200	100	100	Knowledge Management Competencies in Finance Course	0.84	0.022	0.83	0.022
Behforouz and Al Ghaithi (2024)	Oman	1	2	3	1	60	20	20	Grammar Proficiency - Past Continuous Tense	0.132	0.1	0.13	0.098
	Oman	1	2	3	1	60	20	20	Grammar Proficiency - Past Simple Tense	10.01	0.935	9.81	0.916
Bhatia et al. (2024)	India	1	4	1	1	100	50	50	Knowledge of Acquisition of Anatomical Landmarks	1.01	0.045	0.99	0.045
Boudouaia et al. (2024)	Algeria	1	4	3	1	76	39	37	EFL Reading Comprehension	1.65	0.062	1.63	0.06
Chang et al. (2024)	Taiwan	1	4	2	1	50	25	25	Critical Thinking	1.79	0.11	1.75	0.14
		1	4	2	1	50	25	25	Problem-solving	1.2	0.09	1.19	0.09
Chang et al. (2025)	Taiwan	1	4	2	1	44	22	22	Higher Order Thinking Skills	−1.25	0.109	−1.23	0.107
Chen and Gong (2025)	China	2	2	2	1	50	25	25	Academic Writing	0.78	0.086	0.77	0.084
Chen et al. (2025a)	China	2	4	2	1	64	32	32	Critical Thinking	3.26	0.146	3.22	0.144
Chen et al. (2025b)	China	1	3	2	1	64	32	32	Design history knowledge	0.95	0.07	0.93	0.068
Çiçek et al. (2025)	Türkiye	1	4	1	3	115	59	56	Clinical reasoning skills	0.21	0.035	0.21	0.035
Dasari et al. (2024)	Indonesia	1	3	4	1	20	11	9	Mathematical performance	0.78	0.217	0.75	0.208
de la Puente et al. (2024)	Colombia	1	4	4	3	95	47	48	Argumentation	0.95	0.047	0.94	0.046
		1	4	4	3	95	47	48	Complex Concepts	0.77	0.045	0.76	0.045
		1	4	4	3	95	47	48	Critical Thinking	0.74	0.045	0.73	0.045
Fardian et al. (2025)	Indonesia	2	4	1	1	22	10	12	Linear Algebra Learning	1.19	0.216	1.13	0.207
Feng and Wang (2024)	China	1	2	3	1	83	42	41	English - CET 4	2.28	0.08	2.26	0.079
		1	2	3	1	83	42	41	English - CET 6	2.03	0.073	2.01	0.072
		1	2	3	1	83	42	41	English - Final Exam	1.48	0.061	1.46	0.061
Gan et al. (2024)	China	1	4	2	1	110	56	54	Final exam: Obstetrics & Gynecology	0.39	0.037	0.39	0.037

(continued on next page)

Table 2 (*continued*)

Study	Country	Study Type	Exposure	Majors	PAGCA	N	N_1	N_2	Learning Outcome	d	var_d	g	var_g
		1	4	2	1	110	56	54	Final exam: Surgery	0.45	0.037	0.44	0.037
		1	4	2	1	110	56	54	Orthopedics MCQ Test: A1 Items Correct	0.49	0.038	0.48	0.037
		1	4	2	1	110	56	54	Orthopedics MCQ Test: A2 Items Correct	0.38	0.037	0.38	0.037
		1	4	2	1	110	56	54	Orthopedics MCQ Test: A3/4 Items Correct	0.62	0.038	0.61	0.038
		1	4	2	1	110	56	54	Orthopedics MCQ Test: Total Correct	0.29	0.037	0.29	0.036
Gasaymeh and AlMohtadi (2024)	Jordan	1	4	2	3	74	36	38	Visual Basic Programming Language	0.48	0.056	0.47	0.055
Guo et al. (2024)	China	1	4	2	1	124	60	64	Writing Ability: Content	0.67	0.034	0.67	0.034
		1	4	2	1	124	60	64	Writing Ability: Language	0.38	0.033	0.38	0.033
		1	4	2	1	124	60	64	Writing Ability: Organization	0.15	0.032	0.15	0.032
		1	4	2	1	124	60	64	Writing Ability: Total score	0.52	0.033	0.51	0.033
Habib et al. (2024)	USA	2	4	4	2	75	75	75	Creativity: Elaboration	0.54	0.01	0.54	0.03
		2	4	4	2	75	75	75	Creativity: Flexibility	0.84	0.02	0.83	0.03
		2	4	4	2	75	75	75	Creativity: Fluency	0.79	0.02	0.78	0.03
		2	4	4	2	75	75	75	Creativity: Originality	0.62	0.02	0.62	0.03
Hakiki et al. (2023)	Indonesia	1	4	1	3	62	31	31	Student Engagement in Technology Learning	1.43	0.09	1.41	0.08
He and Lu (2024)	China	1	1	2	1	60	30	30	Achievement: A2 (ChatGPT) vs A3 (Open-Book)	0.839	0.073	0.828	0.071
		1	1	2	1	60	30	30	Achievement: A2 (ChatGPT) vs B2 (No ChatGPT)	1.284	0.081	1.268	0.079
Hongxia and Razali (2025)	China	2	2	2	1	70	36	34	Overall English Academic Writing	0.95	0.064	0.94	0.063
Hu et al. (2025)	China	1	2	2	1	53	26	27	Art Appreciation Capability	3.09	0.169	3.04	0.164
Huang et al. (2025)	China	1	4	2	1	187	93	94	Operation Performance	0.67	0.023	0.67	0.022
Ji et al. (2023)	China	1	2	1	1	36	15	21	Learning performance – Fluency	1.98	0.172	1.94	0.165
		1	2	1	1	36	15	21	Learning performance – Novelty	1.85	0.165	1.81	0.158
		1	2	1	1	36	15	21	Learning performance – Originality	0.64	0.12	0.62	0.115
		1	2	1	1	36	15	21	Learning performance - Total	1.9	0.167	1.85	0.16
		1	2	1	1	36	15	21	STEM Teaching Competence	−0.36	0.116	−0.35	0.111
		1	2	1	1	36	15	21	STEM Teaching Knowledge	1.63	0.153	1.59	0.146
		1	2	1	1	36	15	21	STEM Teaching Literacy	0.01	0.114	0.01	0.109
Kavadella et al. (2024)	Cyprus	2	3	2	1	70	31	39	Knowledge Exam (Radiation Biology & Protection)	0.52	0.06	0.51	0.059
*Kim (2024)	South Korea	2	3	4	1	50	50	50	English Writing Performance	1.32	0.057	1.31	0.057
Kofahi and Husain (2025)	Jordan	1	4	4	3	53	23	30	Higher-Order Thinking Ability	0.62	0.08	0.61	0.079
Kosar et al. (2024)	Slovenia	1	4	4	3	182	89	93	Programming Performance	−0.03	0.022	−0.03	0.022
Kucuk (2024)	Iraq	2	2	2	1	60	30	30	Grammar Proficiency	1.93	0.099	1.91	0.096
Lee et al. (2024)	China	1	2	1	1	61	30	31	Chemistry Knowledge Test	0.63	0.067	0.62	0.065
Li (2023)	China	1	2	4	1	81	39	42	Creative Thinking	0.48	0.05	0.48	0.05
		1	2	4	1	81	39	42	Students' Project Performance on Designing Courseware	0.91	0.05	0.9	0.05
Li (2023b)	China	1	4	2	1	43	21	22	Chinese Writing Test	0.76	0.1	0.75	0.097
Li (2025)	China	1	2	4	1	300	150	150	Creativity: Abstractness	2.72	0.026	2.71	0.026
		1	2	4	1	300	150	150	Creativity: Elaboration	2.63	0.025	2.62	0.025
		1	2	4	1	300	150	150	Creativity: Fluency	2.73	0.026	2.72	0.026
		1	2	4	1	300	150	150	Creativity: Originality	2.81	0.027	2.8	0.026
		1	2	4	1	300	150	150	Creativity: Resistance to Premature Closure	2.41	0.023	2.41	0.023
Li et al. (2025)	China	2	3	4	1	61	31	30	Content	0.87	0.072	0.86	0.07
		2	3	4	1	61	31	30	Language	0.72	0.07	0.71	0.068
		2	3	4	1	61	31	30	Overall Writing Score	0.63	0.069	0.62	0.067
		2	3	4	1	61	31	30	Structure	0.05	0.066	0.05	0.064
Liu et al. (2025a)	China	2	2	2	1	91	45	46	Academic Writing	0.97	0.046	0.96	0.045
Lu et al. (2024)	China	1	4	2	1	48	24	24	Achievement Score	0.63	0.087	0.62	0.086
Luan et al. (2025)	China	2	4	3	3	106	106	106	Long-Term Retention	−4.22	0.103	−4.04	0.103
Mahapatra (2024)	India	1	2	1	1	72	35	37	Academic Writing	1.331	0.068	1.316	0.067
		1	2	1	1	72	35	37	Academic Writing	2.21	0.09	2.186	0.089
Moheno et al. (2024)	Mexico	1	3	2	1	75	25	50	Frequency of Use of ChatGPT	−0.412	0.061	−0.41	0.061
		1	3	2	1	75	25	50	Knowledge Level of ChatGPT	−0.285	0.061	−0.28	0.06
		1	3	2	1	75	25	50	Pedagogical Knowledge	0.378	0.061	0.37	0.06

(*continued on next page*)

Table 2 (*continued*)

Study	Country	Study Type	Exposure	Majors	PAGCA	N	N_1	N_2	Learning Outcome	d	var_d	g	var_g
Niloy et al. (2024)	Bangladesh	1	3	3	1	600	300	300	Content Presentability	0.309	0.007	0.308	0.007
		1	3	3	1	600	300	300	Elaboration/Logic Development of Ideas	0.293	0.007	0.293	0.007
		1	3	3	1	600	300	300	Total Creativity Score	−0.731	0.007	−0.73	0.007
		1	3	3	1	600	300	300	Writing Content Accuracy	−1.127	0.008	−1.126	0.008
Qin et al. (2024)	China	2	3	2	1	102	51	51	Course Scores - Lab	1.172	0.046	1.163	0.046
		2	3	2	1	102	51	51	Course Scores - Theoretical	1.461	0.05	1.45	0.049
		2	3	2	1	102	51	51	Course Scores - Total	1.471	0.05	1.46	0.049
Qureshi (2023)	Saudi Arabia	1	1	1	2	24	12	12	Problem-solving	1.22	0.19	1.18	0.2
Seo (2024)	South Korea	1	3	3	1	44	44	44	Narrative Writing skill	0.843	0.054	0.835	0.053
Smailova et al. (2025)	Kazakhstan	1	1	2	1	52	25	27	Intermedial Writing Proficiency - Balance & Focus	0.127	0.077	0.125	0.076
		1	1	2	1	52	25	27	Intermedial Writing Proficiency - Integration	0.142	0.077	0.14	0.076
		1	1	2	1	52	25	27	Intermedial Writing Proficiency - Relevance & Insights	0.238	0.078	0.235	0.076
		1	1	2	1	52	25	27	Intermedial Writing Proficiency - Specificity & Depth	0.19	0.077	0.188	0.076
Song and Song (2023)	China	2	2	3	1	50	25	25	Language Use	0.89	0.09	1.19	0.09
		2	2	3	1	50	25	25	Overall Writing Proficiency	0.93	0.09	0.92	0.09
		2	2	3	1	50	25	25	Writing Content	0.65	0.08	0.87	0.09
		2	2	3	1	50	25	25	Writing Organization	0.82	0.08	1.09	0.09
Sun et al. (2024)	China	2	4	2	2	82	39	43	Program Performance Score	0.31	0.05	0.31	0.05
Uddin et al. (2023)	USA	1	2	1	3	42	42	42	Hazard Recognition Ability	3.44	0.17	3.38	0.12
Wahba et al. (2024)	Jordan	1	2	2	1	56	28	28	Statistical Reasoning	0.95	0.079	0.93	0.078
Wang (2025)	China	1	2	1	1	68	35	33	English Communication Ability	3.5	0.178	3.48	0.174
		1	2	1	1	68	35	33	English Communication Ability (Phase 2)	5.22	0.329	5.19	0.322
Wang and Feng (2023)	China	1	4	2	2	83	42	41	Reading Comprehension	2.37	0.08	2.35	0.08
		1	4	2	2	83	42	41	Writing Book Review	3.2	0.11	3.17	0.11
Yang et al. (2024)	Taiwan	1	4	2	1	71	36	35	Python Learning Achievement	0.63	0.051	0.62	0.05
Yıldız (2023)	Türkiye	2	4	3	3	60	30	30	Overall Ability	5.57	0.3	5.5	0.297
Yilmaz and Yilmaz (2023)	Türkiye	1	4	1	2	45	24	21	Computational Thinking Skills	1.12	0.1	1.1	0.1
Yusof (2025)	Malaysia	1	3	2	1	128	64	64	Final Exam Score in Measurement & Evaluation in Education	0.645	0.033	0.641	0.033
Zhang et al. (2024)	China	1	4	1	1	34	17	17	Retention Test	−0.05	0.118	−0.05	0.115
		1	4	1	1	34	17	17	Transfer Test	−0.55	0.122	−0.54	0.119
Zhang et al. (2025)	China	1	2	3	1	64	31	33	Reading Achievement	1.28	0.075	1.26	0.074
Zhong et al. (2023)	Singapore	2	4	3	1	130	30	100	Disciplinary Grounding	0.77	0.04	0.77	0.04
		2	4	3	1	130	30	100	Integration	0.82	0.04	0.82	0.04
		2	4	3	1	130	30	100	Interdisciplinary Learning Diversity	0.76	0.04	0.76	0.04
		2	4	3	1	130	30	100	Problem Solving	0.59	0.04	0.59	0.04
Zhou and Kim (2024)	South Korea	1	2	2	1	74	38	36	Music Learning Outcomes	0.41	0.028	0.41	0.028
Zhu et al. (2023)	Singapore	2	4	3	1	130	72	58	Collaborative Interdisciplinary Problem-solving Skills	0.37	0.03	0.36	0.03

Note. 1.The positive effect size indicates ChatGPT's great effect on undergraduate students' learning outcomes.
2. For Study Types, 1 = Quantitative Study; 2 = Mixed Methods Study.
3. Exposure indicates students' previous exposure to ChatGPT usage, where 1 = all participants have previous exposure to ChatGPT usage; 2 = None of the participants have previous exposure; 3 = some of the participants have previous exposure; 4 = not specified.
4. Majors: 1 = STEM majors; 2 = non-STEM majors; 3 = some participants are STEM majors, and some are non-STEM majors; 4 = Not specified.
5. PAGCA indicates Presence or Absence of Guidance in the ChatGPT Application, where 1 = Presence of guidance in ChatGPT application; 2 = Absence of guidance in ChatGPT application; 3 = Not specified.
6. N is the total sample size. N1 represents the number of students in the control group for a control-experimental design study or the number of students in the pre-intervention group in a repeated measure design study. N2 represents the number of students in the experiment group in a control-experimental design study or the number of students in the post-intervention group in a repeated measure design study.

7. d represents Cohen's d, the standardized mean difference. vard is the variance of Cohen's d. g denotes Hedges' *g*, a bias-corrected effect size. var_g is the variance of Hedges' *g*.

3.4. Studies with multiple effect sizes

Some studies included in our meta-analysis (Chang et al., 2024; Habib et al., 2024; Song & Song, 2023; Wang and Feng 2023; Zhong et al., 2023) provided multiple effect sizes within the same study, which can lead to data dependency when using effect sizes from the same sample (Strube, 1987; Wood, 2008). We handled dependent effect sizes by applying Robust Variance Estimation (RVE) in our statistical analysis. Using RVE is a well-supported and pragmatic choice for handling multiple effect sizes from the same study (Hedges et al., 2010) because its application provides a practical approach to handling dependent effect sizes by producing valid standard errors and test statistics without requiring assumptions about the exact dependence structure (Hedges et al., 2010). To assess the robustness, we also examined how the results are stable across the range of within-study correlation values (ρ) from 0 to 0.9 in RVE analyses.

Additionally, some effect sizes were excluded from the respective studies because they were not relevant to our objective of analyzing undergraduate students' learning outcomes. For example, in the study by Yilmaz and Yilmaz (2023), we retained the effect size for computational thinking skills but excluded the effect sizes for programming self-efficacy and learning motivation. In the study by Song and Song (2023), we retained the effect sizes of overall writing proficiency and excluded those of writing motivation. Similarly, for the study by Chang et al. (2024), we kept the effect sizes of problem-solving skills and critical thinking skills, while excluding those of learning enjoyment.

3.5. Transformation to a common metric across different study designs

The selected articles varied significantly in their methodologies and outcomes. Seven studies (Almineeai et al., 2025; Alwasidi & Al-Khalifah, 2025; Habib et al., 2024; Luan et al., 2025; Seo, 2024; Uddin et al., 2023) employed a pre-/post intervention repeated-measure design, while the remaining 61 studies utilized an independent control/experimental design. According to Ray and Shadish (1996), adjustment is necessary for making direct comparisons or aggregation among effect sizes from various experimental designs (Glass et al., 1981; Morris & DeShon, 1997). We first calculated the effect sizes using Cohen's *d,* representing the pre-post standardized mean differences for the two repeated-measure design studies then converted the effect sizes from repeated measures to the effect size with a common metric using the following formula proposed by Morris and DeShon (2002): $d_{IG} = d_{RM}\sqrt{2(1-p)}$, where d_{IG} refers to the targeted independent group effect size, d_{RM} is the repeated measure effect sizes that needs to be converted, and p is the pre-intervention and post intervention correlation, calculated using the following formula (Morris & DeShon, 2002): $r = \frac{SD_{pre}^2+SD_{post}^2-SD_D^2}{(2)(SD_{pre})(SD_{post})}$. This approach ensures the effect sizes from repeated-measure studies being comparable with those from the independent groups design studies. The effect size is further transformed to Hedges' *g* for meta-analysis.

3.6. Data analysis

We applied the random-effects model as we expected effect sizes to differ across studies (Borenstein et al., 2021; Hunter & Schmidt, 2004). We first visualized variation of the effect sizes across studies with a forest plot (see Fig. 3). We then computed the overall average effect size across the 129 effect sizes using inverse-variance weighting based on their standard errors. After computing the weighted average of retrieved effect sizes, we ran Cochran's *Q* test for heterogeneity in effect sizes among studies and computed the percentage of variation across studies (I^2) and the between-study variance estimate (τ^2) (Higgins & Thompson, 2002). We examined publication bias using both a visual inspection of the funnel plot (see Fig. 3) and Egger's test for funnel plot asymmetry (Lin & Chu, 2018).

We further explored effect size variation across studies using four potential moderators: (1) study types, (2) previous exposure to ChatGPT, (3) students' majors, and (4) the presence or absence of guidance in ChatGPT application. For each moderator, we employed RVE with small-sample corrections, specifying a within-study correlation of $\rho = 0.8$. This approach accounts for dependencies among effect sizes from the same study. To statistically test the significance of each moderator, we performed an omnibus *F* test by fitting meta-regression model. All analyses were performed using the *Metafor* (version 4.6; Wolfgang, 2023) and *Robumeta* (version 2.1; Fisher et al., 2023) packages in R, within the R Studio Integrated Development Environment (IDE; version 2024.04.2).

4. Results

4.1. Heterogeneity of effect sizes

Table 2 reports the Hedges' *g* and its standard error for each study. Under the RVE model (with setting $\rho = .8$), the weighted average effect size (Hedges' *g*) was 1.14 with *SE* of 0.185 ($n = 66$ studies, 129 outcomes, 95 % *CI* [0.77, 1.51], $p < .01$). This indicates that ChatGPT use has a substantial positive impact on undergraduate students' learning outcomes, exceeding Cohen's conventional threshold for large effects ($g > 0.8$). Sensitivity analyses demonstrated the robustness of this finding, with effect size estimates remaining stable ($g \approx 1.14$) across different within-study correlation assumptions ($\rho = 0$ to 0.9).

Fig. 2 presents a forest plot displaying the distribution of effect sizes across the 66 included studies, which visualizes the heterogeneity in ChatGPT's impact on learning outcomes among 129 individual effect sizes. The overall weighted average effect size ($g = 1.14$) is represented by the diamond at the bottom of the plot. The distribution reveals considerable variability in both the magnitude and precision of effects across studies, with Hedges' *g* effect sizes ranging from −4.04 to 11.69. Aligning with the observation on the forest plot, the heterogeneity test indicated that significant variation exists among the effect sizes ($Q(128) = 2959.03$, $\tau^2 = 1.92$, $I^2 = 97.62\%$, $p < .01$). Although the *Q* test is likely to yield significance with a large sample, the forest plot and high heterogeneity indices together support the presence of systematic between-study variance contributing to the observed heterogeneity in effect sizes.

4.2. Publication bias

A funnel plot was utilized to assess the publication bias among the included studies. Fig. 3 depicts the majority of effect sizes cluster near the center of the plot, but with noticeable asymmetry. The result of Egger's test for funnel plot asymmetry also indicates evidence of publication bias ($z = 9.06$, $p < .001$) among the included studies. As depicted in Fig. 3, those studies with small sample sizes (i.e., wider *CI*s) tend to report disproportionally large positive effects, while more precise studies with large primary sample sizes tend to report smaller or more neutral effects. Although the two large positive effects (Ali et al., 2025; Behforouz & Al Ghaithi, 2024) located at the right bottom of the chart carry smaller weights for synthesis and are therefore less influential in estimating the overall effect, their presence still poses a risk of overestimating the average effect. This observation led us to move on

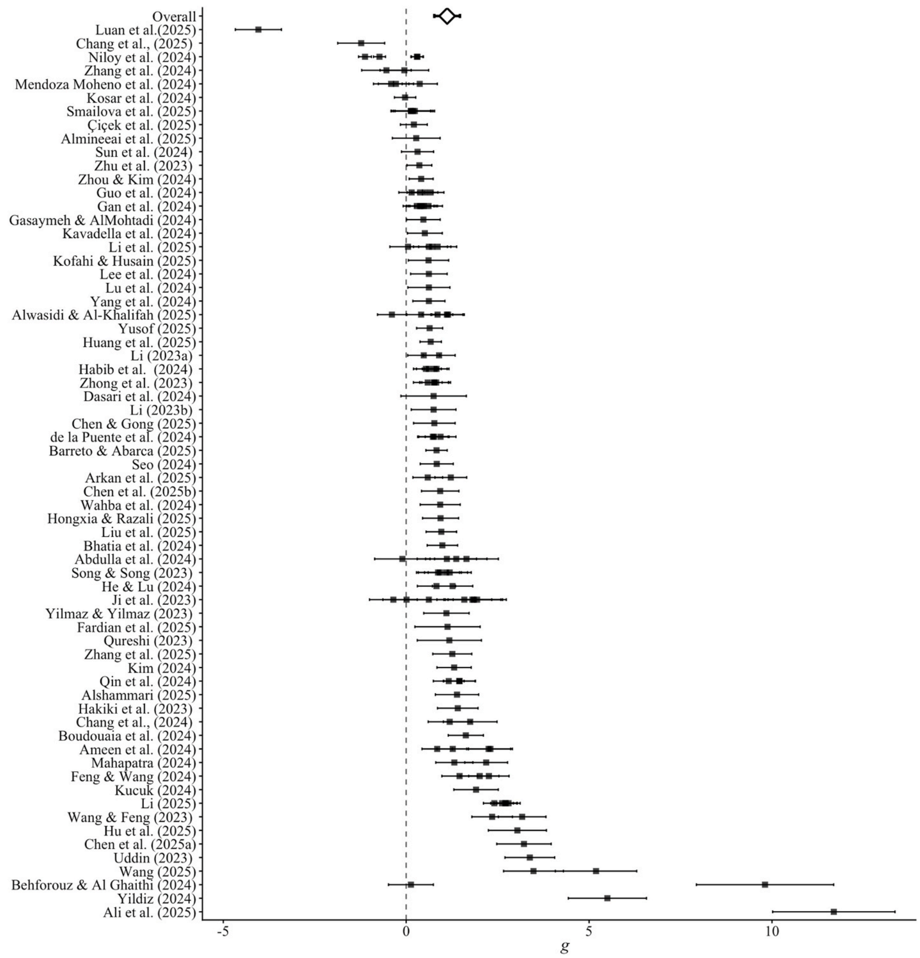

Fig. 2. Forest plot of effect sizes retrieved from 66 studies.

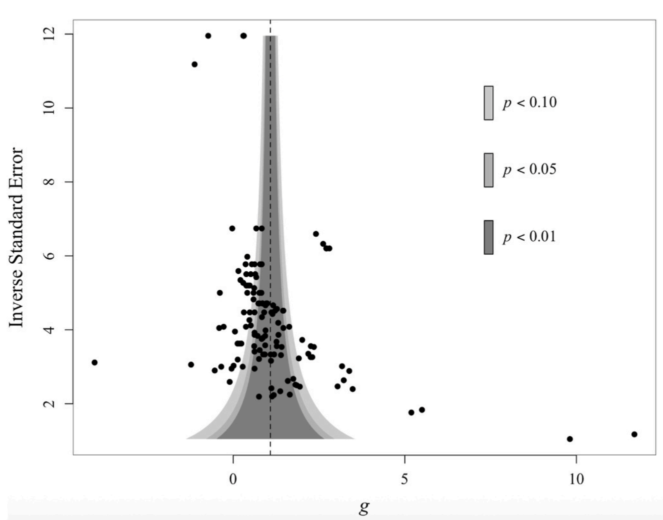

Fig. 3. Funnel plot.

investigating the potential moderators with caution to explain the variation among the effect sizes and further scrutinize design elements of each of the included studies to better understand the effective education intervention with ChatGPT.

4.3. Moderator analysis

To further explore sources of heterogeneity in effect sizes, RVE-based moderator analyses were conducted across four variables: Study Type, Previous Exposure to ChatGPT, Majors, and Presence or Absence of Guidance in the ChatGPT Application (See Table 3). The significance of each moderator was assessed using omnibus *F* tests derived from Wald tests.

4.4. Study type

The omnibus F test indicated no evidence of an overall moderating effect for study type ($F = 0.18$, $df = 35.5$, $p = .68$). Quantitative studies produced a large and statistically significant effect ($g = 1.07$, 95 % *CI* [0.76, 1.38], $p < .01$), suggesting that quantitative research tends to report a strong positive effect of ChatGPT's application on undergraduate students' learning outcomes. Although the quantitative component in Mixed-Methods studies seems to yield slightly larger effects, on average, this difference was not statistically significant ($\beta_g = 0.23$, $SE = 0.54$). Thus, regardless of the study type or methodological design, the impact of the ChatGPT remains consistently large and positive across studies.

4.5. Previous exposure to ChatGPT

The significant omnibus F test ($F = 3.59$, $df = 16.5$, $p = .04$) indicates that exposure levels meaningfully moderated the variability in the effect sizes across included studies. This suggests that the degree of exposure

Table 3
Moderator analysis.

Moderators	Moderator Levels	Number of ES (n)	Estimate	SE	t	95 % *CI* Lower	95 % *CI* Upper	Omnibus ***F***	I^2 (%)	τ^2
Study Type	Quantitative+	92	1.07	0.16	6.87**	0.76	1.38	0.18	95.43	1.15
	Mixed Methods	37	0.23	0.54	0.42	−0.86	1.31			
Exposure	None+	37	1.73	0.29	6.07**	1.13	2.34	3.59*	94.83	1.06
	All	9	0.57	1.62	0.35	−3.38	4.52			
	Some	27	−1.07	0.32	−3.34**	−1.73	−0.41			
	Not Specified	56	−0.93	0.38	−2.47*	−1.70	−0.17			
Majors	STEM+	23	1.81	0.61	2.94*	0.48	3.13	0.66	95.28	1.16
	Non-STEM	57	−0.89	0.63	−1.41	−2.19	0.41			
	Mixed Majors	27	−0.65	0.89	−0.73	−2.48	1.18			
	Not Specified	22	−0.91	0.66	−1.36	−2.30	0.49			
PAGCA	Presence+	102	1.20	0.20	6.05**	0.80	1.60	0.12	95.43	1.16
	Absence	16	−0.13	0.36	−0.36	−0.96	0.70			
	Not Specified	11	−0.36	0.83	−0.43	−2.19	1.47			

Note. 1. Moderator levels marked with+are the reference groups. 2. Significant codes: p-value <0.01 **; <0.05 *.

meaningfully influences ChatGPT's impact on learning outcomes. In studies where participants had no previous exposure to ChatGPT (None), the mean effect size was large and statistically significant ($g = 1.73$, 95 % *CI* [1.13, 2.33], $p < .01$), indicating that first-time exposure to ChatGPT produced substantial gains in learning outcomes. While studies in which all participants had prior exposure (ALL) seem to show a larger average effect size, the mean difference was not statistically significant, possibly due to the large variation among the effect sizes within the category ($\beta_g = .57, SE = 1.62$). In contrast, studies involving participants with mixed exposure level (Some) demonstrated a significantly lower effect size ($\beta_g = -1.07, SE = .32,\ p < .01$), suggesting diminished learning gains when participants varied in their familiarity with ChatGPT. Similarly, studies that did not specify exposure level (Not-specified) reported a significant but lower average effect ($\beta_g = -.93$, $SE = .38$), further indicating that the effect of ChatGPT seems to be largest when participants had the first exposure to the tool and the lack of clarity or inconsistency in prior exposure is associated with reduced learning benefits.

4.6. Majors

The omnibus F test ($F = 0.66$, $df = 26.2$, $p = .58$) indicates that academic major did not significantly moderate the overall variability in effect sizes across studies. For STEM majors, the mean effect was large and statistically significant ($g = 1.81$, 95 % *CI* [0.48, 3.13], $p < .01$), indicating that the application of ChatGPT yielded substantial learning gains in STEM-related domain. In contrast, the coefficients for Non-STEM majors ($\beta_g = -.89, SE = .63$), Mixed majors ($\beta_g = -.65, SE = .89$), and studies in which the academic major was not specified (Not Specified , $\beta_g = -.91, SE = .66$) indicated smaller average effects. However, none of these differences relative to STEM reached statistical significance, likely due to large *SE*s. Although this pattern suggests a potential trend favoring stronger benefits within STEM disciplines, the current evidence is insufficient to confirm systematic differences across disciplines.

4.7. Presence or absence of guidance in the ChatGPT application

The omnibus F test was not significant ($F = 0.13$, $df = 12.2$, $p = .88$), indicating that the presence of guidance in ChatGPT applications did not significantly moderate the variability across studies. When the presence of guidance was explicitly reported, the mean effect was large and statistically significant ($g = 1.20$, 95 % *CI* [0.80, 1.60], $p < .01$), suggesting that guided use of ChatGPT may support improved learning outcomes. In contrast, studies reporting the absence of the guideline (Absence, $\beta_g = -.13, SE = .36$) or not specifying its presence (Not Specified ,$\beta_g = -.36, SE = .83$) showed smaller average effects; however, these differences are not statistically significant. It is worth noting that the majority of included effect sizes ($n = 102$) reported providing guidance for ChatGPT usage, while a smaller number of studies reported either no guidance or did not specify whether guidance was provided ($n = 16$ and 11, respectively). The wide confidence intervals associated with these subgroups further limit the ability to draw firm conclusions regarding the moderating role of guidance in ChatGPT implementation.

In summary, results indicate that exposure to ChatGPT significantly moderated the overall heterogeneity in effect sizes across the included studies, suggesting that the degree of exposure meaningfully influences ChatGPT's impact on learning outcomes. In contrast, study type, academic major, and the presence of guidance in the ChatGPT applications did not significantly moderate effect sizes. Despite these subgroup differences based on the exposure levels accounted for some variability, between-effect size heterogeneity remained substantial ($I^2 > 94$ %). This suggested that the moderator did not fully explain the observed heterogeneity, and additional, untested moderators may contribute to the variation in effect sizes.

5. Discussion

The rapid growth of AI in education, especially through large language models like ChatGPT, is transforming learning (Ghnemat et al., 2022). As universities adopt these technologies, assessing their impact on undergraduate students' learning outcomes is crucial. It is essential to evaluate how these tools enhance or impede the learning process, contribute to different learning styles, and address the diverse needs of students. Thus, our research aims to contribute to the evolving landscape of undergraduate education in the era of AI.

Our findings in this meta-analysis indicate a substantial positive impact of ChatGPT on undergraduate student learning across diverse disciplines, with the overall weighted average effect size ($g = 1.14$). The large effect size suggests that ChatGPT can be a highly effective aid in educational settings, providing students with valuable assistance in understanding and applying knowledge. It also highlights its promise to be a powerful tool for fostering academic growth and innovation. These findings align with previous research demonstrating the positive impact of AI tools on education (Chen et al., 2023; Hakiki et al., 2023; Kim et al., 2022).

5.1. Comparative analysis of existing meta-analyses on ChatGPT and student learning

Several meta-analyses have examined the influence of ChatGPT on student learning, but they vary considerably in scope, methodological rigor, and findings. He and Cordie (2024) synthesized nine studies ($n = 9$) and reported a moderate effect size (Cohen's $d = 0.60$), but their work did not extend to moderator analyses, limiting its explanatory depth. Similarly, Liu, Zuo, and Lu (2025) reviewed 37 studies ($n = 37$) and found a pooled Hedges' $g = 0.577$, indicating a moderate positive effect of ChatGPT application on student learning. Their subgroup analyses suggested stronger effects in social sciences, medium-length interventions, and declarative knowledge tasks, though they cautioned about potential inflation of effect sizes due to the treatment of multiple outcomes as independent.

Meanwhile, Chen and Xu (2024) adopted a narrower scope with 28 studies ($n = 28$) and identified a small overall effect size ($g = 0.254$). Their investigation into moderators such as learning environment, school level, and type of knowledge added nuance but indicated weaker evidence of ChatGPT's effectiveness compared to other syntheses. In contrast, Wang and Fan (2025) conducted a broader synthesis of 51 studies ($n = 51$), focusing on three outcome categories. They reported a large effect on learning performance ($g = 0.867$). Their inclusion of moderator analyses revealed that course type, instructional model, and intervention duration systematically influenced effect sizes.

Our meta-analysis provides an extensive coverage to date (May 31st, 2025), synthesizing 66 studies with 129 effect sizes and 5708 cases, yielding a much larger pooled effect size (Hedges' $g = 1.14$). Unlike He and Cordie (2024) and Chen and Xu (2024), we conducted moderator analyses to examine contextual factors that might explain heterogeneity in outcomes. These analyses considered variables such as study type, student majors, previous exposure to ChatGPT, and the presence of guidance on ChatGPT applications. Among these, previous exposure to ChatGPT emerged as a significant moderator, indicating that differences in participants' experiences with the tool contributed meaningfully to variations in learning outcomes across studies. Compared with Liu, Zuo, and Lu (2025) and Wang and Fan (2025), our analysis draws on a larger body of evidence, strengthening generalizability while also confirming the importance of moderators in interpreting effect sizes. In summary, these comparisons highlight both the growing consistency in findings that ChatGPT supports student learning and the unique contribution of our study in providing the most comprehensive and fine-grained evidence to date.

5.2. Study type as a lens for implementing ChatGPT in higher education

Our findings from the moderator analysis highlight several important implications for the use of ChatGPT regarding undergraduate students' learning. First, although Study Type did not significantly moderate the overall relationship between ChatGPT applications and undergraduate students' learning outcomes, the observed pattern of stronger effects in quantitative studies suggests that ChatGPT's benefits may be most apparent in structured, evidence-based contexts. This trend offers meaningful guidance for the intentional and data-driven integration of AI tools such as ChatGPT in higher education.

According to the Integrative AI in Education (IAIE) Framework, at the learner level, this finding suggests that ChatGPT may influence learning outcomes differently based on the types of tasks or assessments being evaluated. Quantitative studies, which often measure concrete performance indicators such as test scores, assignment quality, or accuracy, tend to capture measurable and objective learning outcomes, highlighting ChatGPT's application in structured, performance-oriented contexts. This pattern aligns with personalized learning, where AI tools support learners in achieving specific, measurable goals and provide immediate, data-driven feedback (Hew et al., 2023).

For educators, the contrast between quantitative studies and mixed-methods studies implies that ChatGPT may be better suited for certain teaching strategies. For example, it could enhance learning outcomes in structured, data-driven environments (as seen in quantitative studies) rather than in less structured, qualitative contexts. This allows educators to tailor their use of ChatGPT for tasks where AI's strengths, such as providing formative feedback or supporting skill development, are most beneficial (Hooda et al., 2022; Lee & Moore, 2024). Additionally, the impact of study type suggests that AI integration strategies should be evidence-based, recognizing that ChatGPT might yield more tangible benefits in learning environments where outcomes are measurable.

At the institutional level, policies could prioritize ChatGPT integration in domains where learning outcomes can be systematically and quantitatively assessed, as the observed trend toward stronger effects in quantitative studies suggests that ChatGPT's benefits are more easily demonstrated in data-driven, evidence-based contexts. Institutions could further support this by promoting research that uses measurable indicators of learning success while complementing them with qualitative insights to capture broader educational impacts of ChatGPT and other AI tools.

5.3. Disciplinary difference in ChatGPT integration

The moderator analysis indicated that Major did not significantly moderate the relationship between ChatGPT and undergraduate students' learning outcomes. However, the stronger effects observed among STEM major students suggest the possible disciplinary variations in how students engage with and benefit from ChatGPT integrated learning environments. At the learner level, the significant positive effects for STEM majors suggest that ChatGPT applications may align well with structured and analytical characteristics of STEM education. ChatGPT applications could support students through structured problem solving, personal assistance, and feedback, which yield measurable improvements in learning outcomes. At the instructor level, STEM educators can integrate ChatGPT into well-structured tasks, which makes ChatGPT's impact more observable and statistically robust. At the institutional level, STEM programs may have infrastructures for data-driven teaching and assessment, providing systemic reinforcement that amplifies ChatGPT's effectiveness.

In contrast, the non-significant effects of ChatGPT applications on students with non-STEM majors, mixed majors, or majors not specified indicate that ChatGPT has less effect on these disciplines. At the learner level, open-ended and interpretive tasks in non-STEM disciplines do not align well with ChatGPT's structured output (Wu & Zhang, 2023). At the educator level, non-STEM educators may have fewer standardized entry points for AI integration, making the instructional impact less observable (Khadivi, 2023; Parviz, 2024). At the institutional level, less structured infrastructure or policy support for AI integration in these disciplines may further reduce the measurable results (Varghese et al., 2025; Wu & Zhang, 2023).

This finding has important implications for practice. At the learner level, AI systems could be adapted for non-STEM domains by incorporating discipline-specific tasks, enabling students in the humanities and social sciences to use AI tools such as ChatGPT as a partner for interpretation, reflection, and creativity rather than providing structured answers. At the instructor level, professional development is essential for instructors in non-STEM disciplines to identify meaningful entries for AI integration through model argumentation, generation of alternative perspectives, or scaffolding critical thinking, given standardized applications are less available compared to STEM fields. At institutional level, universities could design policies, infrastructure, and funding mechanisms that support non-STEM adoption, including pilot projects and targeted investments to expand AI use beyond STEM fields. To achieve equitable and effective benefits across disciplines, AI integrations must be discipline-specific and supported by both pedagogical support and institutional commitment. In this way, students in all majors will have opportunities to benefit meaningfully from AI-enhanced learning.

5.4. Leveraging guidance to enhance ChatGPT's impact on learning

The moderator analysis reveals that the Presence of Guidance in ChatGPT Applications did not significantly moderate the overall learning outcomes. However, studies explicitly reporting guidance showed large and significant effects, while those coded as Absence or Not Specified yielded relatively smaller, non-significant effects. This indicates that the way ChatGPT is scaffolded and contextualized in instruction may meaningfully shape its educational impact. The results highlight that guided use of ChatGPT enhances students' ability to use the tool productively and efficiently.

At learner level, guidance in ChatGPT applications provides students with structured prompts, scaffolds, and clear directions for how to use ChatGPT effectively. Without this support, learners may use ChatGPT inconsistently or in unproductive ways, leading to weaker or non-significant outcomes. The significant results when guidance is present highlight that AI alone is insufficient and the learner's experience must be intentionally guided. Additionally, these findings highlight that instructors play a critical role in embedding guidance into ChatGPT integration in education. By actively mediating AI use through structured prompts, scaffolds, and clear directions, educators can magnify the learning benefits of AI tools for students. At the institutional-system level, institutions should develop policies and training programs that encourage guided use of ChatGPT in classrooms. This may involve providing educators with resources on best practices for structuring AI tasks or embedding guidance features into institutional AI platforms. By institutionalizing guided practices, the benefits of AI integrations can be scaled more consistently across fields.

5.5. *Exposure is not enough: Structured integration matters more than familiarity*

The moderator analysis for Previous Exposure to ChatGPT revealed a significant moderating effect, indicating that students' prior experience with ChatGPT meaningfully influenced their learning outcomes. Contrary to expectations, studies where students had no previous exposure to ChatGPT yielded large and significant learning gains. In contrast, studies where students had full, partial, or unspecified prior exposure showed either weaker or non-significant effects. This pattern suggests that structured, first-time exposure within a guided ChatGPT application may be more impactful than unstructured or accumulated prior use.

At the learner level, the results suggest that students with structured and uniform exposure to ChatGPT tend to benefit most. While the group with full prior exposure demonstrated relatively low estimated effect size ($g = 0.57$), this effect was not statistically significant, likely due to a small number of studies with large variation within the category ($n = 9$). In contrast, students with no prior exposure experienced a large and statistically significant effect ($g = 1.73$), indicating that first-time, guided use of ChatGPT can produce substantial learning gains. In comparison, studies with mixed exposure or unspecified exposure yielded significantly lower learning outcomes, suggesting that inconsistency in prior exposure may hinder the effectiveness of ChatGPT-based interventions. These findings highlight the importance of equitable, intentional onboarding for all students to ensure they start from a common baseline when engaging with AI in learning.

For educators, the results highlight the importance of strategically scaffolding students' exposure to ChatGPT. Educators should avoid assuming that prior experience automatically translates into higher learning outcomes. Instead, educators should implement progressive, differentiated learning activities that accommodate varying levels of prior experiences, while moving all students toward effective use of AI tools. In a mixed-exposure classroom, unequal familiarity can lead to disparities in learning outcomes. Therefore, a deliberate instructional approach that ensures structured exposure, supports novice users, and deepens experienced students' engagement is essential for maximizing the effectiveness of AI tools on students' learning outcomes.

At the institutional level, these findings highlight the need for equitable approaches to integrating AI tools like ChatGPT into curricula. The significant moderating effect of prior exposure suggests that inconsistent or unstructured familiarity with ChatGPT may undermine its educational impact. Institutions should implement baseline AI literacy programs that provide all students with guided, consistent exposure to AI tools like ChatGPT. This approach not only promotes learning equity but also ensures that the benefits of emerging technologies are accessible to all learners. Additionally, institutions must support faculty through professional development and establish clear guidelines for AI use in curricula to ensure system-wide, effective integration of AI tools in education.

In summary, ChatGPT is an adaptable AI tool with the potential to enhance education at the learner, educator, and institutional levels. For learners, it offers an accessible platform that supports personalized learning, enabling students to use the tool effectively for diverse tasks such as language learning, writing, and problem-solving (Kohnke et al., 2023; Orrù et al., 2023). This adaptability empowers students to tailor ChatGPT's use to meet their individual needs, making AI-enhanced learning experiences accessible to a broad range of users (Goel, 2020; Song et al., 2024). For instructors, ChatGPT can increase teaching efficiency and allow them to focus on core instructional goals while supporting students in learning through AI integrations (Tan et al., 2025). At the institutional level, ChatGPT's versatility enables efficient AI integration across departments and programs, promoting broad applicability across disciplines, allowing institutions to deploy ChatGPT widely with optimized resource allocation. This promotes equitable access to advanced learning technologies. ChatGPT's adaptability and ease of use make it an invaluable tool at every level of the educational system, facilitating the seamless integration of AI to enhance learning, streamline instruction, and support institutional innovation in education.

6. Limitations and future directions

While this meta-analysis provides valuable insights into the impact of ChatGPT on undergraduate students' learning experiences, several limitations warrant consideration. One potential limitation is the relatively small number of moderators we explored —Study Type, Students' Majors, Previous Exposure to ChatGPT, and Presence or Absence of Guidance in ChatGPT Application. This limited scope may exclude critical variables that could reveal differential impacts of ChatGPT on undergraduate student learning. Consequently, replications with diverse learning contexts and broader sets of moderators are essential to strengthen or refine the current findings. Future research should aim to produce more robust and generalizable conclusions about the educational implications of ChatGPT, examining how contextual factors, learner characteristics, and instructional designs jointly influence its effectiveness.

In addition, publication bias remains a potential limitation of this meta-analysis study. It is likely that studies reporting significant or favorable results were more readily published, particularly given the growing interest in AI tools in education. Consequently, the pooled effect size may be somewhat inflated. Future syntheses should incorporate more unpublished or gray literature to obtain more balanced estimates.

Finally, the rapid evolution of AI technology presents both opportunities and challenges for maintaining the relevance of these findings. The current findings are based on studies published between January 1st, 2023, and May 31st, 2025. As new evidence emerges, the significance of these findings may be challenged. Thus, we encourage researchers to continue exploring ChatGPT's applications in education, particularly its long-term effects on learning outcomes as the technology matures. Investigating how educators adapt their instructional strategies to leverage ChatGPT's evolving features will also provide valuable guidance for its effective integration in education contexts.

By addressing these potential limitations, future studies can enhance the robustness and comprehensiveness of findings in this rapidly evolving field. Scholars have just started generating important evidence-based knowledge in this area, and ongoing research will play a pivotal role in advancing our understanding of the broader implications of AI technologies like ChatGPT in educational contexts.

7. Conclusion

The rise in research on ChatGPT's role in education requires a comprehensive review to assess its strengths and weaknesses. As additional studies emerge, it becomes increasingly crucial to continuously update this knowledge, ensuring the effective and responsible adoption of AI technologies such as ChatGPT. This meta-analysis lays the groundwork for future ongoing efforts to accumulate research evidence, providing valuable insights and recommendations for educators, researchers, and policymakers in higher education. These efforts will assist in optimizing the use of ChatGPT as a supportive educational tool, enhancing undergraduate students' learning and addressing potential challenges.

CRediT authorship contribution statement

Fangfang Mo: Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Jing Huang:** Writing – review & editing, Visualization, Methodology, Data curation. **Yao Yang:** Writing – original draft, Investigation, Data curation. **Zafer Özen:** Data curation. **Yukiko Maeda:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis. **F. Richard Olenchak:** Writing – original draft.

Statement on open data, ethical considerations, and conflict of interest

This meta-analysis study was conducted exclusively using data extracted from previously published, peer-reviewed studies. No new data were collected, and no experiments involving human or animal subjects were carried out. Therefore, approval from an institutional review board or ethics committee was not required.

All data analyzed in this study are contained within the original publications cited in the reference list. The dataset compiled for this meta-analysis, including extracted summary statistics and calculated effect sizes, is presented in the tables within the manuscript, in accordance with open science and data sharing principles.

There is no potential conflict of interest in this study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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